

YUNSOO KIM | University College London

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EDUCATION

University College London, London, UK

PhD in Health Informatics

Sep. 2023 – May. 2026 (expected)

Thesis: "Foundation Model for Collating Clinical Expert Knowledge: Facilitating Histopathology and Radiology with Multimodal LLMs"

Supervisors: Professor Honghan Wu, Dr Adam Levine, Professor Robert Stewart, Dr Nikolas Pontikos

Imperial College London, London, UK

M.Sc. in Bioinformatics and Theoretical Systems Biology

Oct. 2015 - Sep. 2016

Thesis: "[A Clustering-based Short Time Series Analysis Methodology in Omics](#)"

Supervisors: Professor Tim Ebbels, Dr John Pinney, Dr Virginia Fairclough

- Developed a novel network inference methods ([differential](#) and [co-occurrence](#))

Case Western Reserve University, Cleveland, OH

B.S. with Honors in Systems Biology (Concentration in Bioinformatics, Pre-Health Plan)

Sep. 2012 - May. 2015

Honors Thesis: "Assembly of Marama chloroplast from Next-gen sequencing reads"

Supervisor: Professor Christopher Cullis

- First author paper at the Journal of Experimental Botany in 2016
[A novel inversion in the chloroplast genome of marama \(Tylosema esculentum\)](#)

RESEARCH INTEREST

Computational biologist and AI researcher specializing in multimodal learning for clinical applications. Experienced in leading cross-functional teams and delivering impactful research at the interface of deep learning, medical imaging, and bioinformatics. Passionate about bridging academia and industry to advance biomedical AI.

EMPLOYMENT

LG CHEM, Seoul, South Korea

Jul. 2017 – Sep. 2023

Professional (Principal Researcher) Mar. 2022 – Sep. 2023

Specialist (Senior Researcher) Mar. 2019 – Feb. 2022

Associate (Researcher) Jul. 2017 – Feb. 2019

NLP Group Leader, CTO Artificial Intelligence Project

April. 2020 – Sep. 2023

- Led NLP research group (a group of 6 researchers), developing advanced language models for scientific discovery and R&D acceleration
- [Developed a BERT-based artificial reading system for research candidate recommendation, successfully adopted in multiple domains \(battery, petrochemical, biopolymer\)](#)
- Drove adoption of NLP-driven solutions across interdisciplinary research teams, enhancing productivity and discovery

LG AI Research

LG AI Master Program Pilot, Representative Researcher

Sep. 2019 – Jan. 2020

- Represented LG Chem in a cross-divisional AI leadership program
- Improved BERT model in biotechnology, especially in Agricultural Biology – Achieved SOTA in 7 BioNLP benchmark datasets

CTO Green Bio Project at LG Chem

Jul. 2017 – Mar. 2020

- Developed plant disease recognition and plant disease severity diagnosis with ResNext, EfficientNet ensemble models
- Developed an early allergen and toxin assessment system and [a neural network for predicting protein allergenicity/toxicity, improving screening efficiency and safety](#) (First patent).

Military Manpower Administration, South Korea

Aug. 2017 – Aug. 2020

Technical Research Personnel, Alternative military service at LG Chem

- Fulfilled mandatory service by contributing as a full-time researcher, advancing AI and biotech projects to support Korea's scientific and technological development.

Torrey Pines Co., Ltd., Cheonan, Korea

Oct. 2016 – July. 2017

Co-Founder (-Sep. 2023) and CEO.

- [Launched 'Bota nouveau', Korea's one of the first microbiome-focused hair care cosmetics brand](#)

SELECTED PUBLICATIONS

Look & Mark: Leveraging Radiologist Eye Fixations and Bounding boxes in Multimodal LLMs for CXR Report Generation	Conference: ACL 2025 Findings
First Author	
BioHopR: A Benchmark for Multi-Hop, Multi-Answer Reasoning in Biomedical Domain	Conference: ACL 2025 Findings
First Author	
IHC-LLMiner: Automated extraction of tumour immunohistochemical profiles from PubMed abstracts using LLMs	Under review
First Author	
Enhancing Human-Computer Interaction in Chest X-ray Analysis using Vision and Language Model with Eye Gaze Patterns	Conference: MICCAI 2024
First Author	
MedExQA: Medical Question Answering Benchmark with Multiple Explanations	Conference: ACL 2024 BioNLP Workshop
First Author	
All you need is continued pretraining of chemistry texts even for molecule captioning	Conference: ACL 2024 Language Plus Molecule Workshop
First Author	
Foundation Model for Biomedical Graphs: Integrating Knowledge Graphs and Protein Structures to Large Language Models	Conference: ACL 2024 Student Research Workshop
First Author	
Human-in-the-Loop Chest X-Ray Diagnosis: Enhancing Large Multimodal Models with Eye Fixation Inputs	Conference: IJCAI 2024 TAI4H Workshop
First Author	2024
Hallucination Benchmark in Medical Visual Question Answering	Conference: ICLR 2024 Tiny Paper (Notable Accept)
Co-first Author	2023
Chemical Language Understanding Benchmark	Conference: ACL2023 Industry Track
First Author	

Journal Reviewing

IEEE Transactions on Medical Imaging, BMC Medical Informatics and Decision Making

Conference Reviewing

ICLR2025, MICCAI2025, HealTac2025, ACL2025 ARR

SELECTED PATENTS

Literature Chemical Entity Recognition Method and System through Adjusting Sentence Length in Training Data	2023
Main Inventor	Patent Application: KR20240171755A
AI-based Chemical Entity Recognition System and Method	2023
Main Inventor	Patent Application: KR20240171756A
Document Classification System and Method Using Entity Group Scores	2022
Main Inventor	Patent Application: KR20240168118A
Method to build chemical language model and NLP document analysis system and method applying it	2021
Main Inventor	Patent Application: KR20230037335A
Chemical Named Entity Recognition Method and System through Data Preprocessing	2021
Main Inventor	Patent Application: KR20230030337A
Protein Toxicity Prediction System and Method Using Artificial Neural Network	2019
Main Inventor	Patent Granted: KR102757409B1

SELECTED HONORS AND AWARDS

UCL Faculty of Population Health Sciences PGR Travel Fund, £650 (~\$900)	2024 - 2025
FACT Graduate Scholarship, £5,000 (~\$7,000)	2024
ACL SRW Grant, \$1,050	2024
Rosetrees Trust Grant, £36,000 (~\$49,000)	2024
LG AI IDEathon Digital Transformation Frontier (Applications 1st place; Overall 2nd place)	2021
LG Chem Seoul National University AI Hackathon 1st place	2021
LG AI Advanced problem-solving School Best Practice Mentor, ~\$3,000	2020
LG AWARDS (Best Award given by LG Group Chairman), ~\$12,500	2020
Multiple research and innovation awards at LG, including:	2019 – 2022
- AI/DX Best Practice Award (awarded 3 times)	
- AI Organization Manager Award (awarded 2 times)	
- Best Research Report (awarded 2 times)	
- CEO Excellent R&D Project Award	
- On-Spot Incentive (awarded 3 times)	
Case Western Reserve University Scholarship, \$40,000	2012 – 2015
Ogelbay Summer Research Grant, ~\$5,000	2014

OTHER EXPERIENCE

Teaching Experience, Workshops, Seminars

MICCAI2025 Tutorial Main Organizer and Speaker	2025
MICCAI2025 Workshop Main Organizer and Speaker	2025
HealTac2025 Tutorial Chair	2024 – 2025
UCL Institute of Ophthalmology Guest Lecture on LLMs in Medicine	2024
MICCAI2024 Tutorial Main Organizer and Speaker	2024
HealTac2024 Tutorial Main Organizer and Speaker	2024
UCL MSc student co-supervisor for research project	2023 – 2024
LG Chem Community of Practice for ChatGPT main organizer	2022 – 2023
LG Chem graph neural network model study group lead organizer	2022 – 2023
LG Chem transformer model (NLP) study group main organizer	2021 – 2023
LG Chem Community of Practice for Systems Metabolic Engineering AI main organizer	2022
LG AI Advanced problem-solving School Mentor	2020
LG Chem Internal Python Instructor	2020
LG AI weekly Study Group member, organizer	2019 – 2022
LG Chem Bio Big Data and AI Community of Practice member, organizer	2017 – 2019

Invited Talks, Presentation

European Congress of Pathology Oral Presentation	2025
HealTAC Poster Presentation and Demo	2025
RECOMB Poster Presentation, Lightning Talk	2025
MICCAI Poster Presentation	2024
ACL BioNLP, LPM Workshops Poster Presentation	2024
ACL Student Research Workshop Poster Presentation	2024
IJCAI TAI4H Oral Presentation	2024
Lightning Talk at AI in Pathology: Research Meets Reality at the Alan Turing Institute	2024
UCL Computational Pathology Meetings Invited Talk	2024
HealTAC PhD Thesis Oral Presentation and Poster Presentation	2024
ICLR Tiny Paper Poster Presentation	2024
University College London Institute of Health Informatics Talk on Large Language Model Research in Medicine	2024
University of Edinburgh Clinical NLP Group Talk on Large Language Model Research in Medicine	2023
ACL2023 Industry Track Poster Presentation	2023
LG Chem CTO Digital Transformer talks	2022
LG Chem Platform Technology Workshop talks	2022
LG Chem Platform Technology Artificial Intelligence annual meeting speaker	2021
LG Chem Tech Fair talks	2021
LG DX Fair 2021 speaker	2021
LG Chem Digital Transformation talks	2021
LG Chem CTO Digital Transformer talks speaker	2021
LG Chem Platform Technology Workshop talks speaker	2021
LG Chem Digital Transformation Best Practice talks	2020
LG Academy Digital Transformation talks panelist	2020
LG Housys AI Master talks speaker	2019
LG AI Master talks speaker	2019
LG Chem Green Biology quarterly meeting speaker	2019